

Seungju Han

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PROFESSIONAL EXPERIENCE

ViFiVE, New York, NY

Aug 2022 – May 2025

Data Scientist

- Deployed and optimized machine learning models for on-device production in iOS and web environments, ensuring high performance and reliability.
- Boosted **3D pose estimation accuracy by 17%** with a custom self-attention module, improving clinical assessment performance.
- Developed an comprehensive ML pipeline using MySQL, Firestore, and Python, from data parsing to model production, **boosting clinical assessment efficiency by 30%** across multiple tests.
- Revamped data augmentation pipeline for a human pose estimation model, **achieving a 10% increase in computational efficiency**, reducing model training time, and accelerating overall project timeline.
- **Enhanced a Random Forest pose classification model accuracy from 89% to 96%** by applying advanced feature engineering techniques, boosting reliability of clinical assessment predictions.
- Led efforts to validate the proficiency of AI-based models in clinical assessments with researchers at Stanford and Seoul National University Hospital, successfully applying vision AI models to high-stakes assessments like the Short Physical Performance Battery test, **impacting over 300 patients and clinicians** across different regions.

New York Mets, Remote

Oct 2021 – Jul 2022

Data Scientist

- Generated expected distribution model for infield and outfield batted balls with 89% accuracy by leveraging pitcher and batter data, including pitch types, pitcher/batter handedness, and batted ball statistics.
- Applied K-Means and KNN clustering methods to group similar pitchers, simulating more realistic pitcher-batter matchups. This approach led to a **25% improvement in prediction accuracy** in player performance forecasts.
- Leveraged data analysis predictive models to inform defensive alignment strategies, which contributed to the Mets **improving their Defensive Runs Saved (DRS) by 130%**, from +20 DRS in 2021 to +46 DRS in 2022.

Fastcampus Language, Seoul, Korea

Jun 2020 – Aug 2020

Business Analyst

- Automated KPI dashboard for leadership team, **reducing operating time by 75%** through integrating Google Suite tools (BigQuery, Looker, and Google Sheets).
- Leveraged SQL to analyze data for multiple teams, uncovering key trends and providing actionable insights that improved decision-making and contributed to **13% decrease in operating cost on average**.

PROJECTS

Pose Human Pose Estimation with Dinov2

Github: https://github.com/seungjoohan/dino_pose

- Implemented and explored the application of a Vision Transformer (ViT) model for 3D human pose estimation, leveraging LoRA (Low-Rank Adaptation) for fine-tuning.
- Developed a data processing pipeline for handling pose datasets and integrated augmentation strategies.

Scalable Movie Rating Prediction

- Developed a scalable end-to-end data pipeline to preprocess over 28 million movie and user records using Spark and stored engineered features in MongoDB for efficient retrieval.
- Trained an Alternative Least Square (ALS) collaborative filtering model, achieving r2 score of 0.83 within 2 hours using distributed machine learning, leveraging the scalability of Spark to process the large dataset efficiently.

EDUCATION

University of San Francisco, San Francisco, CA

Jul 2022

MS in Data Science

Relevant Coursework: Machine Learning, Deep Learning, Linear Regression, Time Series Analysis, A/B Testing, Relational Databases, NoSQL, Distributed Computing (Spark), Data Structures & Algorithms

University of California Berkeley, Berkeley, CA

Dec 2019

BA in Statistics

TECHNICAL SKILLS

Python (numpy, pandas), Machine Learning (pytorch, tensorflow, scikit-learn), Deep Learning, SQL (BigQuery), NoSQL (MongoDB, Firestore), Spark, Data Visualization (matplotlib, Seaborn), AWS (S3, EC2, EMR), Databricks, GCP